



Pimpri Chinchwad Education Trust's
Pimpri Chinchwad College of Engineering (PCCoE)
(An Autonomous Institute)
Affiliated to Savitribai Phule Pune University (SPPU)

MDM-Digital Marketing - Capstone Project [BCE27MD06]

Department: Computer Engineering (RL) **Academic Year:** 2026 - 2027 **Semester:** I
Year: B. Tech **Group ID:** Group 9 **Date:** 22/08/2026

Problem Statement:

Customer Journey Intelligence: An AI-Driven, Multidisciplinary Framework for Personalised Digital Marketing

SDG Goals aligned with the project:

SDG No.	Name
SDG 8	Decent Work and Economic Growth
SDG 9	Industry, Innovation and Infrastructure

Team Members:

PRN No.	Name of Student	Department	Mail-id	Contact No.	Signature
123B1B180	Lokeek Kishor Lokhande	COMP	lokeek.lokhande23@pccoepune.org	7558527810	
123B1B041	Gauri Avinash Dhondge	COMP	gauri.dhondge23@pccoepune.org	9689467489	
123B1B017	Ved Keshav Mahajan	COMP	ved.mahajan23@pccoepune.org	8421074696	

Project Guide: Dr. Rachana Patil

1. Introduction

This project is set in the e-commerce/retail digital marketing domain where customers generate behavioural data across many touchpoints — advertisements, website visits, product searches, product views, carts and post-purchase interactions — before a purchase decision is made.

Digital marketing's role has shifted from one-time, mass-broadcast campaigns toward continuous, data-driven engagement that adapts to how a customer actually behaves; recent comprehensive reviews of AI in digital marketing confirm that predictive analytics and machine learning are increasingly central to this shift toward adaptive, customer-responsive strategy [1].

The motivation for this project is that most existing analytics systems still answer narrow, isolated questions — click-through probability, purchase probability, or which single product to recommend — rather than modelling the customer's journey as a connected sequence; recent data-driven journey-optimisation research shows that integrating behavioural signals into a single adaptive framework can meaningfully improve predictive accuracy over such isolated, rule-based models [2].

Current digital marketing practice commonly relies on funnel dashboards, RFM-style customer segments, and big-data-driven targeting rules to personalise campaigns [3], [4]. While effective for broad targeting, these practices generally stop short of building an explicit, chronological journey for each customer or explaining why a specific marketing action is recommended at a specific moment — the gap this project is motivated to address, including formally supporting "no action" as a valid marketing decision.

2. Literature / Market Review

Customer segmentation remains one of the most active areas of digital-marketing research: a structured review of segmentation methods for e-commerce customer targeting finds that RFM-based representation combined with clustering (e.g., K-Means) is the most frequently used and best-performing approach across the reviewed literature, though it identifies open questions around deep-learning-based segmentation [3]. Complementing this, strategic-marketing research shows that firms which integrate big-data analytics into a dynamic capabilities model achieve more adaptive, evidence-driven marketing decisions than firms relying on static segmentation alone [4]. Machine-learning-based analysis of large-scale behavioural/social data has similarly been shown to reveal decision-making patterns that improve targeted engagement, for example in customer decision-making analysis for the hospitality sector [5].

On the explainability side, a recent framework integrating SHAP-based explainability with quality-function-deployment for transitioning from Customer Relationship Management (CRM) to Customer Journey Management (CJM) demonstrates that tree-based models paired with Shapley explanations can identify the key behavioural determinants of engagement and purchase, and translate them into operationally actionable plans [6] — directly supporting this project's explainable Next Best Marketing Action layer. Dynamic, individual-level personalisation is further supported by counterfactual-analysis research showing that modelling customer-behaviour transitions (rather than static segment membership) improves the granularity of personalised marketing decisions [7], while industry-oriented studies confirm that combining segmentation with machine learning improves personalised-marketing outcomes in applied settings such as financial services [8].

On the predictive side, tree-based gradient-boosting models such as XGBoost have been shown to outperform traditional classifiers for user purchase-behaviour prediction on real e-commerce interaction data [9], and clickstream-based machine-learning models have been developed specifically to predict anonymous online customers' purchase intention from session-level browsing behaviour [10]. On the personalisation-and-engagement side, recent consumer-behaviour research shows that AI-driven personalisation measurably shapes trust, perceived usefulness and engagement in social-media marketing contexts [11]. Overall, the literature offers strong individual pieces — segmentation methods, explainable prediction, and channel-level personalisation — but no single reviewed system combines chronological journey construction, stage/segment classification, sequence-aware prediction and an explainable, marketer-facing Next Best Action layer with a formal "no action" option, evaluated on open behavioural data; this is the gap this project addresses.

3. Gap Identification

Sr. No.	Existing Approach / System	Limitation Identified	How This Project Addresses It
1	RFM/clustering-based segmentation methods reviewed across e-commerce literature [3]	Segments customers on static behavioural summaries; does not construct an explicit chronological journey per customer	Builds a session-aware, chronological Customer Journey Builder feeding segmentation, not just static RFM scores
2	Big-data-driven strategic marketing / dynamic-capabilities models [4]	Operates at firm/strategy level; lacks a concrete per-customer decision engine	Translates strategy-level capability into a per-customer, rule-based Next Best Marketing Action engine
3	SHAP/QFD-based explainable CRM-to-CJM frameworks [6]	Explains engagement/purchase drivers but stops at analysis; does not formally output a suppress/wait decision	Extends explainable prediction into an action layer that explicitly models Wait and Suppress-Marketing as valid actions
4	Counterfactual/dynamic customer-behaviour segmentation research [7]	Improves personalisation granularity but is not evaluated against alternative marketing decision strategies	Runs an offline strategy-comparison experiment (generic vs. segment-based vs. journey-intelligence) (RQ4)
5	Purchase-prediction models on clickstream/XGBoost data [9], [10]	Optimise prediction accuracy (purchase/intent) only; stop short of a marketer-facing recommendation	Extends prediction models into an explainable Next Best Marketing Action layer with feature-based reasoning

4. Problem Statement

Digital marketing platforms generate large volumes of interaction data (views, searches, cart actions, transactions), yet conventional predictive systems analyse isolated events (e.g., purchase probability) rather than the customer's sequential journey [2], [3]. There is a need for an integrated system that

converts historical interaction data into chronological customer journeys, classifies journey stage and segment, predicts the likely next action and purchase propensity, and recommends an explainable Next Best Marketing Action — including no action — for Admins, Marketing Analysts and Marketing Managers, who are the target users of this system.

5. Objectives

1. To clean, preprocess and construct chronological, session-aware customer journeys from historical interaction data with engineered behavioural features (recency, frequency, view-to-cart ratio, etc.).
2. To classify customers into journey stages (Initial Engagement, Exploration, Consideration, High Intent, Conversion, Retention, Inactive) and behavioural/RFM/ML-based segments.
3. To predict the next likely customer event and purchase propensity, comparing a baseline (Markov chain), classical ML (Logistic Regression, Random Forest, XGBoost) and a sequence-aware model (LSTM/GRU).
4. To design an explainable Next Best Marketing Action engine that recommends actions (cart reminder, email, discount, cross-sell, wait, suppress) with feature-based reasoning.
5. To build journey and marketing analytics dashboards (funnel, drop-off, segment and product analytics) usable by digital marketers.
6. To compare a generic, a segment-based, and a journey-intelligence marketing strategy through offline simulation.

Scope of the Project

- 1.**Market and competitor analysis:** the platform is positioned against commercial NBA/CDP tools and recent AI-in-digital-marketing literature [1] to identify a clear differentiation (see Gap Identification, Section 3), rather than a live competitor-tracking module.
- 2.**Customer/persona analysis:** rule-based, RFM and ML-based (K-Means) segmentation of behavioural data [3].
- 3.**Website/behavioural analysis:** session construction, event standardisation, journey timelines and Customer 360° profiles built on the RetailRocket dataset [12].
- 4.**SEO and keyword research:** reviewed conceptually as a complementary acquisition channel; not implemented as a live module, since the RetailRocket dataset captures on-site behaviour only and contains no search-engine or keyword-level data.

5. Marketing strategy design: retargeting, cart-reminder, re-engagement, personalised email/channel/timing and cross-sell recommendation via the NBMA engine, informed by big-data-driven strategic marketing research [4].

6. Social media and campaign-performance analytics: addressed conceptually through literature on AI-driven personalisation in social-media engagement [11]; live social-channel tracking and real campaign-performance data are out of scope, as the dataset does not capture these channels.

7. AI integration: baseline, classical ML and sequence-aware prediction models [9], [10]; K-Means segmentation [3]; collaborative-filtering-based product recommendation; SHAP/rule-based explainability [6].

8. Web analytics and conversion/ROI analysis: executive dashboard, conversion funnel, journey-flow and drop-off analysis, segment and product analytics; offline ROI-style evaluation (targeting precision, coverage, action distribution) rather than live-campaign measurement.

6. Tools and Technologies

Category	Tools / Technologies
Frontend	React (or Next.js), TypeScript, Tailwind CSS, shadcn/ui, Recharts
Backend	Python, FastAPI
Database	PostgreSQL (Supabase / Neon)
Machine Learning	Pandas, NumPy, Scikit-learn, XGBoost/LightGBM; optional PyTorch/TensorFlow (LSTM/GRU); SHAP
Visualization	Recharts, Plotly
Deployment	Vercel (frontend), Render/Railway (backend), Supabase/Neon (database)
Dataset	RetailRocket e-commerce behavioural dataset [12]

7. Proposed Methodology

Raw interaction data from the RetailRocket dataset [12] is validated, standardised to common event types, and grouped into sessions (30-minute inactivity threshold). A Customer Journey Builder assembles chronological sessions into a Customer Profile, Journey Stage and Behavioural Features, which feed an ML engine: segmentation (rule-based/RFM/K-Means [3]), next-event prediction using clickstream-based purchase-intent modelling [10], and purchase propensity using gradient-boosted models such as XGBoost, shown to outperform traditional classifiers on real purchase-behaviour data [9] (baseline vs. classical ML vs. LSTM/GRU, evaluated with precision, recall, F1 and ROC-AUC using chronological train/test splits to avoid leakage).

Model outputs feed a Decision Engine applying explainable rules (e.g., High Intent + Cart + No Purchase → Cart Reminder; Recent Purchase → Wait/Suppress) to select a Next Best Marketing Action, paired with a Personalization Engine for product/channel/timing suggestions — each with a

SHAP-based or rule-based feature explanation, following the explainable CRM-to-CJM design pattern of integrating Shapley explanations with an operational decision layer [6]. Development proceeds through eight phases: research and dataset finalisation; data pipeline; core backend; ML modelling; decision-engine integration; frontend dashboards; strategy-comparison experiments; and documentation.

8. Result and Discussion

Expected outputs include a journey-stage/funnel dashboard, a model-comparison table (accuracy/precision/recall/F1/ROC-AUC across baseline, classical ML [9] and sequence models [10]) with SHAP-based explanations [6], and an offline strategy-comparison table (generic vs. segment-based vs. journey-intelligence strategy) on targeting precision and unnecessary interventions. These results will be discussed against four research questions — whether sequence modelling improves next-event prediction (RQ1), whether behavioural data yields separable segments (RQ2) as suggested by segmentation-method literature [3], whether journey-aware features improve propensity prediction over static/basic features (RQ3), consistent with counterfactual/dynamic-segmentation findings [7], and whether journey-aware rules reduce irrelevant actions (RQ4) — with all figures reported strictly as offline/simulated evaluation, since the dataset contains no live treatment-outcome data.

9. Conclusion

This project proposes an explainable Customer Journey Intelligence platform that models customer behaviour as a sequence, classifies journey stage and segment, predicts next actions and purchase propensity, and recommends a Next Best Marketing Action — including a deliberate no-action option. Scoped to eight core capabilities and evaluated against four explicit research questions, it aims to show how sequential behavioural intelligence can make digital marketing more personalised, explainable and efficient within an academically honest, reproducible framework.

10. References

- [1] C. Ziakis and M. Vlachopoulou, "Artificial Intelligence in Digital Marketing: Insights from a Comprehensive Review," *Information*, vol. 14, no. 12, p. 664, 2023, DOI: <https://doi.org/10.3390/info14120664>
- [2] T.-C. Wang, R.-S. Guo, C. Chen, and C.-K. Li, "Multi-Stage Data-Driven Framework for Customer Journey Optimization and Operational Resilience," *Mathematics*, vol. 13, no. 7, p. 1145, 2025, DOI: <https://doi.org/10.3390/math13071145>
- [3] M. Alves Gomes and T. Meisen, "A Review on Customer Segmentation Methods for Personalized Customer Targeting in E-Commerce Use Cases," *Information Systems and e-Business Management*, vol. 21, no. 3, pp. 527–570, 2023, DOI: <https://doi.org/10.1007/s10257-023-00640-4>

- [4] C. Brewis, S. Dibb, and M. Meadows, "Leveraging Big Data for Strategic Marketing: A Dynamic Capabilities Model for Incumbent Firms," *Technological Forecasting and Social Change*, vol. 190, p. 122402, 2023, DOI: <https://doi.org/10.1016/j.techfore.2023.122402>
- [5] A. Alsayat, "Customer Decision-Making Analysis Based on Big Social Data Using Machine Learning: A Case Study of Hotels in Mecca," *Neural Computing and Applications*, vol. 35, no. 6, pp. 4701–4722, 2023, DOI: <https://doi.org/10.1007/s00521-022-07992-x>
- [6] T.-C. Wang, "An Explainable AI-Driven Framework Integrating SHAP and QFD for Transitioning from CRM to CJM in Customer-Centric Optimization," *International Journal of Data Science and Analytics*, vol. 22, p. 18, 2026, DOI: <https://doi.org/10.1007/s41060-025-01015-3>
- [7] M. E. Jalal and A. Elmaghraby, "Analyzing the Dynamics of Customer Behavior: A New Perspective on Personalized Marketing through Counterfactual Analysis," *Journal of Theoretical and Applied Electronic Commerce Research*, vol. 19, no. 3, pp. 1660–1681, 2024, DOI: <https://doi.org/10.3390/jtaer19030081>
- [8] C. Potluri, G. Rao, L. V. R. Kumar, K. Allo, Y. Tarekegn, and A. Seman, "Machine Learning-Based Customer Segmentation and Personalised Marketing in Financial Services," in *Proc. 2024 IEEE Int. Conf. on Contemporary Computing and Communications (IC3SE)*, 2024, pp. 1570–1574, DOI: <https://doi.org/10.1109/IC3SE62002.2024.10593143>
- [9] W. Wang, W. Xiong, J. Wang, L. Tao, S. Li, Y. Yi, X. Zou, and C. Li, "A User Purchase Behavior Prediction Method Based on XGBoost," *Electronics*, vol. 12, no. 9, p. 2047, 2023, DOI: <https://doi.org/10.3390/electronics12092047>
- [10] Z. Wen, W. Lin, and H. Liu, "Machine-Learning-Based Approach for Anonymous Online Customer Purchase Intentions Using Clickstream Data," *Systems*, vol. 11, no. 5, p. 255, 2023, DOI: <https://doi.org/10.3390/systems11050255>
- [11] T. Teepapal, "AI-Driven Personalization: Unraveling Consumer Perceptions in Social Media Engagement," *Computers in Human Behavior*, vol. 165, p. 108549, 2025, DOI: <https://doi.org/10.1016/j.chb.2024.108549>
- [12] Retail Rocket LLC, "Retailrocket Recommender System Dataset," Kaggle. [Online]. Available: <https://www.kaggle.com/retailrocket/ecommerce-dataset>

Guide Approval

Project Guide Name: Dr. Rachana Patil

Signature:

Date: